

Markscheme

Mathematics: analysis and approaches
Practice question bank

51. (a)

Consider the equation $2^{x+1} + 2^{2-x} = 9$, where $x > 0$.

Solve the equation for x .

$$\text{let } u = 2^x: 2u + \frac{4}{u} = 9 \quad (M1)$$

$$2u^2 - 9u + 4 = 0 \quad A1$$

$$(2u - 1)(u - 4) = 0 \Rightarrow u = 0.5 \text{ or } u = 4 \quad (M1)$$

$$\text{since } x > 0, u = 2^x > 1, \text{ so reject } u = 0.5; \quad R1$$

$$2^x = 4 \Rightarrow x = 2 \quad A1$$

[5 marks]